$$S = \int d^4x \sqrt{-g} \quad \left[\frac{1}{2} M_{\rm Pl}^2 R - M_{\rm Pl}^2 \left(3H^2(t+\pi) + \dot{H}(t+\pi) \right) + \right. \tag{181}$$
$$+ M_{\rm Pl}^2 \dot{H}(t+\pi) \left((\partial_\mu (t+\pi) \partial_\nu (t+\pi) g^{\mu\nu}) + \right.$$
$$\frac{M_2 (t+\pi)^4}{2!} \left(\partial_\mu (t+\pi) \partial_\nu (t+\pi) g^{\mu\nu} + 1 \right)^2 +$$
$$\left. \left. \frac{M_3 (t+\pi)^4}{3!} \left(\partial_\mu (t+\pi) \partial_\nu (t+\pi) g^{\mu\nu} + 1 \right)^3 + ... \right] \right. ,$$